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Printed Pages – 4 Roll No. 3112413042

324554(24)

B. E. (Fifth Semester) Examination, April-May 2016

(New Scheme)

(Elect. Branch)

INTEGRATED CIRCUITS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Answer all questions. Part (a) from each unit is compulsory and carries 2 marks. Attempt any two parts from part (b), (c) and (d) of each question and carries 7 marks.*

Unit-I

1. (a) Compare ideal and practical OPAMP characteristics. 2

(b) For an op-amp, the input offset voltage is 12 mV, the biasing current is 500 nA and input offset current is 90 nA.

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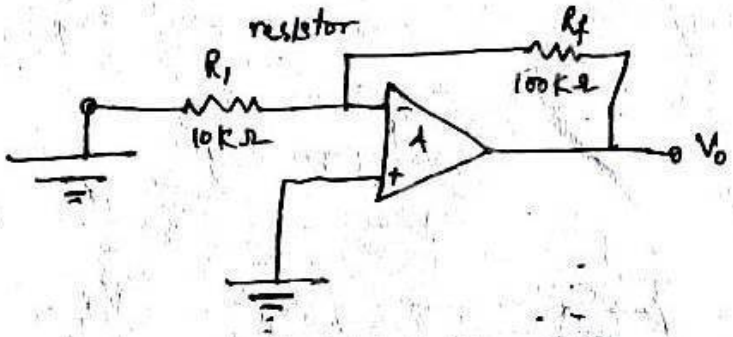
Calculate : 7

(i) Total output offset voltage.

(ii) Compensating resistance required to compensate effect of input bias current.

(iii) Total output voltage with compensating resistor.



(c) Explain briefly the frequency response of OPAMP. 7

(d) For differential amplifier, $R_S = 1\text{K}\Omega$, $R_C = 1\text{K}\Omega$, $h_{fe} = 50$, $h_{ie} = 1\text{K}\Omega$. The emitter resistance is $2{\cdot}5\text{M}\Omega$ while the differential input is 1 mV. Calculate the output voltage and CMRR in dB, if common mode input is 20 mV. Assume single ended output. 7

Unit-II

2. (a) What is level detector? 2

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(b) Draw the circuit diagram and explain commonly used three opamp instrumentation amplifier circuit. Derive expression for its gain. 7

(c) Explain with neat circuit diagrams and waveforms, the working of a negative clamper circuit. 7

(d) Find R_1 & R_f in the loosy integrater so that the peak gain is 40 dB and gain 3 dB down from its peak occurs at a frequency of 1·25 kHz. Use capacitance of 0.01 μF. 7

Unit-III

3. (a) Give the applications of active filters. 2

(b) Explain the operation of 4-bit $R-20\,R$ ladder DAC and derive the expression for output voltage. 7

(c) Design a second order low pass Butterworth filter having high cut off frequency of 1 kHz. Assume $C = 0{\cdot}01$ μF. 7

(d) A Schmitt trigger with hysteresis width 2 V and LTP of −1.5 V converts a 1 kHz sine wave of 5 V peak into a square wave. Calculate the time duration of the negative and positive position of the output wave form. 7

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Unit-IV

4. (a) State any two applications of Astable multivibrator. 2

(b) Design a 555 based square wave generator to produce a symmetrical square wave of 2 kHz. If $V_{CC} = 12$ volts, draw voltage across timing capacitor and the output. 7

(c) Draw the circuit diagram of monostable multivibrator using 555 timer and explain the operation with relevant waveforms. 7

(d) Explain the functional block diagram of IC-555. 7

Unit-V

5. (a) What are the limitations of linear regulator? 2

(b) Compare series and shunt regulator. 7

(c) Design a circuit around 723 to give $V_0 = 10$ V and $I_{L(max)} = 100$ mA. Suggest modifications in the above circuit to handle overcurrent protection. 7

(d) Explain the basic operation of three terminal fixed voltage regulator. 7

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